Diyar Giyash Chapter 9 Research Problem 2

You have been asked to design a network (wired, wireless, or a combination) for a home on which construction will start soon. The home is serviced by a cable TV provider and local phone company, and both provide Internet connectivity. It has two levels totaling 250 meters2 (2700 feet2). Five people will live in the home and use three laptop/tablet computers, a desktop computer, a multimedia server, an Internet-connected TV device (for example, TiVo or digital cable DVR), two multifunction printers, and at least five handheld 802.11n or 802.11ac wireless devices, such as iPods and cell phones. Based on Internet connectivity options in your own area, recommend whether the home should get Internet services via DSL or cable modem. Also, recommend network infrastructure, including a router/firewall, wireless access points, wired switches, and combination devices. Should the homeowner have wired Internet connections installed during construction? If so, what wire types should be used, and in what rooms should connection points be placed?

Internet Service Providers and Network Plan

Rochester is well served and is lucky to have a competitive internet provider market. Spectrum provides internet to around 90% of households with the most basic plans starting at $30/month, and a download speed of up to 1 Gbps. What I recommend for this case, however, is Quantum Fiber. Quantum Fiber offers a fiber optic internet connection that has symmetrical speeds of up to 8 Gbps (symmetrical in this case meaning that the upload and download speeds are the same). A symmetrical speed of 1 Gbps in this case is enough, however. The reason I recommend fiber optic as opposed to copper wiring is because this five person household will be running a multimedia server, multiple 4k video streams, video conferences, remote work, and have multiple phones all connected to the network. This creates a high bandwidth demand, one which fiber optic will be able to fulfill efficiently.

A cheap router can potentially bottleneck the 1 Gbps upload and download speeds of the network, so I would recommend purchasing the Ubiquiti UniFi Dream Machine. This router provides a great firewall, supports a VPN, and can handle the speeds of the fiber optic without slowing it down. For wireless, I would recommend two Wi-Fi 6 (802.11ax) access points, one for each floor of the house. This would cover the 2,700 sq. ft. of the house reliably while providing a consistent and strong connection. WiFi 6 is also backwards compatible with 802.11ac and 802.11n devices, so that won’t be a problem. For switching, I recommend the homeowners purchase a TP-Link

TL-SG1024D 24-port gigabit switch. It would handle all wired connections, leaving the WiFi uncongested and providing a stable connection for stationary devices like printers and desktops. A smaller 8-port switch could also be placed on the second floor if there are enough wired connections that need to be made. This would only be necessary if there are enough devices to justify it. The main cost-saving factor would be that copper wire wouldn't have to be run vertically from floor to floor, saving some money.

Connection Points

Main Floor

The living room should have four Cat6A wall ports. Two on the wall where the multimedia server and T.V will be, one at another convenient location for future connections, and the last one for the ceiling mounted wireless access point. The biggest benefit of these ports will be the direct connection of the multimedia server.

The home office should have three wall ports. One for the desktop, one for the printer, and an extra one for future devices.

The kitchen/dining area could have one port for a smart device such as an Amazon Alexa, but most home owners connect those via WiFi. Considering the heavy WiFi traffic, however, we would want to avoid adding devices to the WiFi as much as possible, so having a port in the dining area for this purpose is advisable if the homeowners are planning on having a smart home device.

Second floor

The master bedroom should have two ports to connect a smart T.V and one extra port for any other device such as a desktop in the future. I would recommend that each of the four additional rooms (for the four other occupants of the home) should have at least one port for stationary devices such as smart T.Vs or desktops. It might be overkill to have two ports for each room, but if the homeowners project each bedroom occupant having more than one stationary device, then two ports per room is an option.

A utility closet would house the 8-port switch I mentioned earlier, but again that would only be necessary if more than one device needs a wired connection per bedroom.

Main Equipment Location

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The main equipment, including the ONT, router, and primary 24-port switch should all be contained in a main floor utility closet or dedicated network panel. Ideally, this closet or panel would be near where the fiber line enters the home for minimal signal loss. Every Cat6A cable originates from this main point, and a single uplink cable connects the secondary 8-port switch on the second floor. This centralizes the equipment and makes troubleshooting, modification, and maintenance much easier and cleaner,

Cost of Equipment and Plan

Quantum Fiber 1 Gbps Fiber Optic Plan	$75/month
TP-Link TL-SG1024D 24-port switch	~$150
Ubiquiti UniFi Dream Machine	~$300
Total	**~$450 + $75/month**